

Nathan O'Brien

nathanobrien.njo@gmail.com

EDUCATION:

Morris County School of Technology, Denville, New Jersey

Class of 2027

Academy for Computer & Information Sciences

GPA (UW): 4.2/4.3

Honor Roll for All MPs Freshman/Sophomore/Junior Year

1530 SAT

All Honors Sophomore/Junior Year

COURSEWORK:

Foundations of Comp Science, Intro to Java, Software Design, Dual enrollment with FDU for Computer Programming I & II, Emerging Technologies, Pre-Calculus, Physics, AP CSP & CSA

EMPLOYMENT:

-Talon Biomarkers

Spring 2026 - Present

**CodeNinjas Sensei
2025**

Spring 2024 - Spring

-US Soccer Licensed Referee

Winter 2023 - Present

PROJECTS:

-Venatio, NASA ADC 2025, Fall 2024 - Winter 2025

National contest where five teammates and I created a simulation for the upcoming Artemis II mission, including antenna connectivity, mission stages, and machine learning.

Photographed in multiple online news sources on the project. Was interviewed by NASA over the project.

JerseyCTF, 5th place, Spring 2026

Placed 5th (student leaderboard) and 9th internationally in an international collegiate cybersecurity competition with over 500 teams participating.

-Eosus: Natural Disaster Awareness, Congressional App Challenge, Summer-Fall 2025

A full-stack, community-driven application created for the Congressional App Challenge using React and Firebase that allows users to report and stay aware of local natural disasters in their area, with maps, AI integration, and more.

-Syracuse Intro to Aerospace Summer Program, Summer 2025

Completed a summer program at Syracuse University, diving into the basics of aerospace engineering. Was able to create multiple lab reports and a portfolio project, as well as a functioning plane wing and glider.

ACTIVITIES:

-FRC Robotics

Fall 2023 - Present

Leader of the Mechanical subteam for MCST Tech Devils (FRC 8588) responsible for CAD and construction of the robot. Attended FRC-licensed events as a drive team member, and helped win the Rising All-Star Award at the Robbinsville competition.

Hack Club Ambassador

Summer 2026 - Present

-Technology Student Association

Fall 2024 - Present

State-level participant for drone design, software development, CAD engineering, and VR visualization. Placed 4th in the state for VR Visualization.

-SkillsUSA

Fall 2024 – Present

State-level participant for the SkillsUSA game design competition for two years.

-Varsity School & Club Soccer

Fall 2023 – Present

Varsity Indoor Track

Winter 2025 - Present

-Volunteer – Pathways Summer Camp Mentor

Summer 2024 – Present

Spent 2 weeks each summer working with elementary-age students with developmental disabilities and teaching them 3D design, Scratch, embedded systems engineering, and more.

-Volunteer – Montville Travel Soccer

Spring 2023 – Present

TECHNICAL SKILLS:

Intermediate Java, Processing Java and Python, HTML/CSS/JS, C#, Unity, Node.js, PostgreSQL, Arduino, and C++

Certified in C# from freecodecamp.com and Microsoft/Codecademy

Certified in Onshape Basics Learning Pathway